Data and Artificial Intelligence
Cyber Shujaa Program

Week 4 Assignment
Business Intelligence on Power BI

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Table of Contents

Data and Artificial Intelligence 1

Cyber Shujaa Program 1

Week 4 Assignment Business Intelligence on Power BI 1

Introduction 3

Objectives 4

Tasks Completed 5

Step 1: Understanding the Hotel Business and Clients' needs 5

Step 2: Loading the Data. 6

Step 3: Transforming the Data. 7

Step 4: Building DAX 8

Step 5: Creating a Visualization Dashboard 10

Link to Dashboard 11

Conclusion 12

Introduction

The week 4 assignment focused on developing hands-on experience in Business Intelligence using Power BI for Hotel Management. It involved loading the hotel datasets into Power BI, transforming the data using Power Query, Building Data Analysis Expressions where I created new columns and measures and finally creating visualizations and coming up with a final dashboard for presentation that highlights the key metrics of the datasets.

Additionally, this assignment has provided me with a more practical approach to understanding how data-driven decision-making can optimize hotel management operations. By working with Power Query, I was able to clean and restructure raw data, ensuring consistency. This transformation step was crucial in preparing the datasets for meaningful analysis, enabling me to extract valuable insights that directly impact business performance.

Building on this foundation, the use of Data Analysis Expressions (DAX) empowered me to create measures and calculations that enhanced the analysis. With well-structured metrics in place, I crafted a compelling visualization dashboard that effectively communicates trends, performance indicators, and business insights. This final step not only honed my ability to present data in an intuitive manner but also reinforced the importance of storytelling through visual analytics in business intelligence.

Objectives

The objectives of the assignment were:

1. Understanding the Hotel Business and Client needs.
2. Loading the Data.
3. Transforming the Data.
4. Building DAX.
5. Creating a Visualization Dashboard.
6. Publishing the Report.

Tasks Completed

Step 1: Understanding the Hotel Business and Clients' needs

Started by first understanding the needs of the hotel business and the clients. Knowing what the client needs in terms of what should be included in the final dashboard, the operations that need to be carried out in order to achieve the final results.

Understanding the hotel business operations like for instance, as per the hostel, Friday and Saturday are considered as weekends while Sunday is considered as a week day.

Understanding the key metrics like;

- Revenue Per Available Room. (RevPAR)
- Average Daily Rate. (ADR)
- Daily sellable rooms.

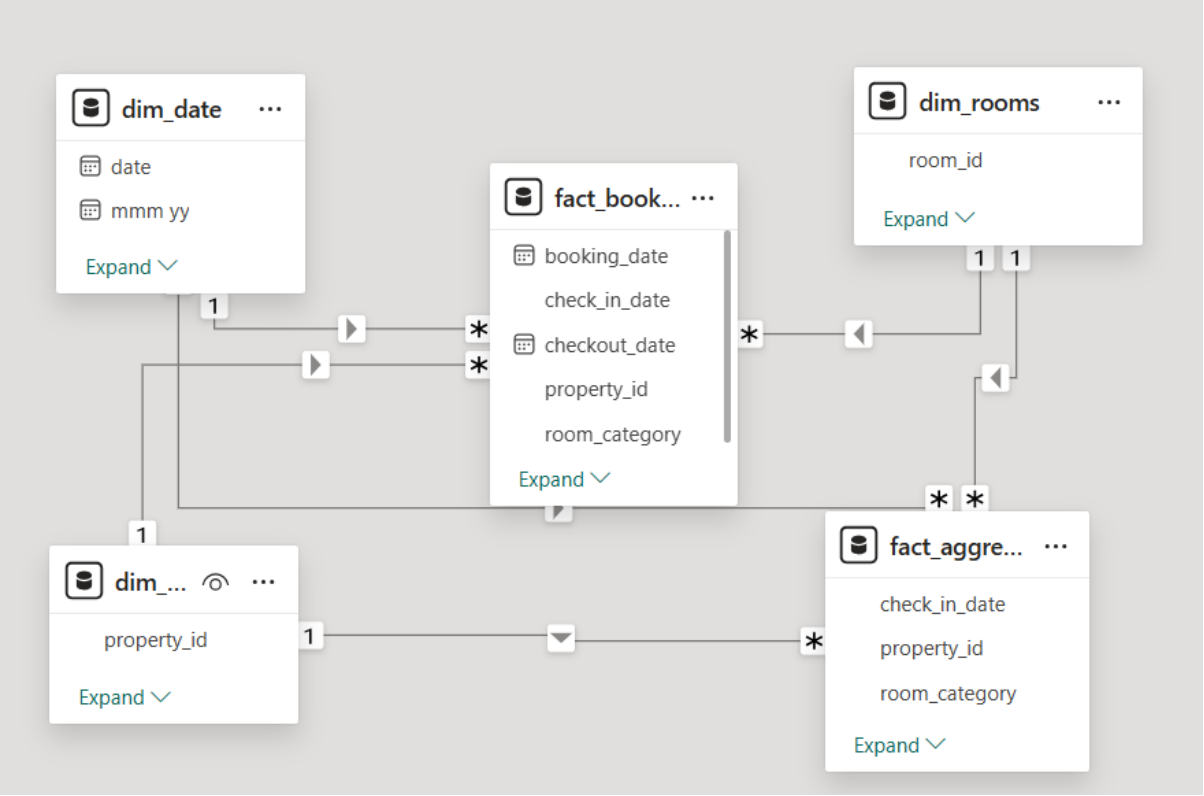
Step 2: Loading the Data.

After understanding what the client needs, it's now time to upload the data to be used.

I stored the files/datasets in a single folder, then loaded the whole folder with the dataset to Power BI instead of loading a single dataset at a time.

Step 3: Transforming the Data.

After loading the data, next step was transforming the data which also involved data modelling. Data Modelling involves creating relationships between the datasets using common or similar columns.
The relationships had a cardinality of one to many.
I applied the Star Schema the facts table needed to be at the middle surrounded by the other tables. For this data, the facts table here was the Fact Bookings Table.



Then did data transformation where I renamed some columns, made sure that that each table had the correct structure for instance making the header rows as the column headers.

Step 4: Building DAX

Creating new Columns.

Data Analysis Expressions involves creating new columns and measures.

Created a new column which has only the week number. Created this new column and derived the week number from the date using the Power BI formulas.

Formula used: ***wn = WEEKNUM(dim_date[date])***

Created another column: ***day_type = WEEKDAY(dim_date[date]).*** This returns the number of days as (1,2,3,4,5,6,7). Friday and Saturday are labeled as 6 and 7 respectively. If the week day is greater than 5, then it should be a weekend. So, Friday and Saturday should be weekends based on the client's needs.

So, to achieve this, I created a variable:

```
day_type =
var wkd = WEEKDAY(dim_date[date])
return if(wkd>5, "Weekend", "Weekday")
```

This formula now returned Friday and Saturday labeled as **"Weekend"** while Sunday as **"Weekday"** in the **day_type** column.

The two columns I created here are the "**wn**" and the "**day_type**" which are referred to as Calculated Columns in DAX.

Creating new a Measure.

Created a new measure named "**key_measures_table**".

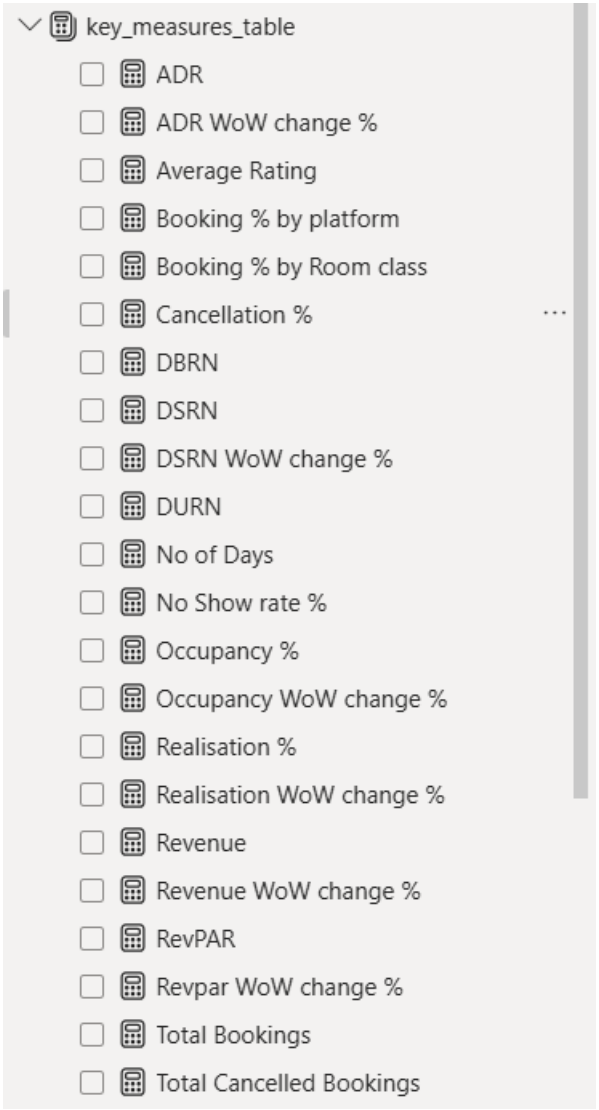
Then created new columns inside the measure using DAX.

For instance:

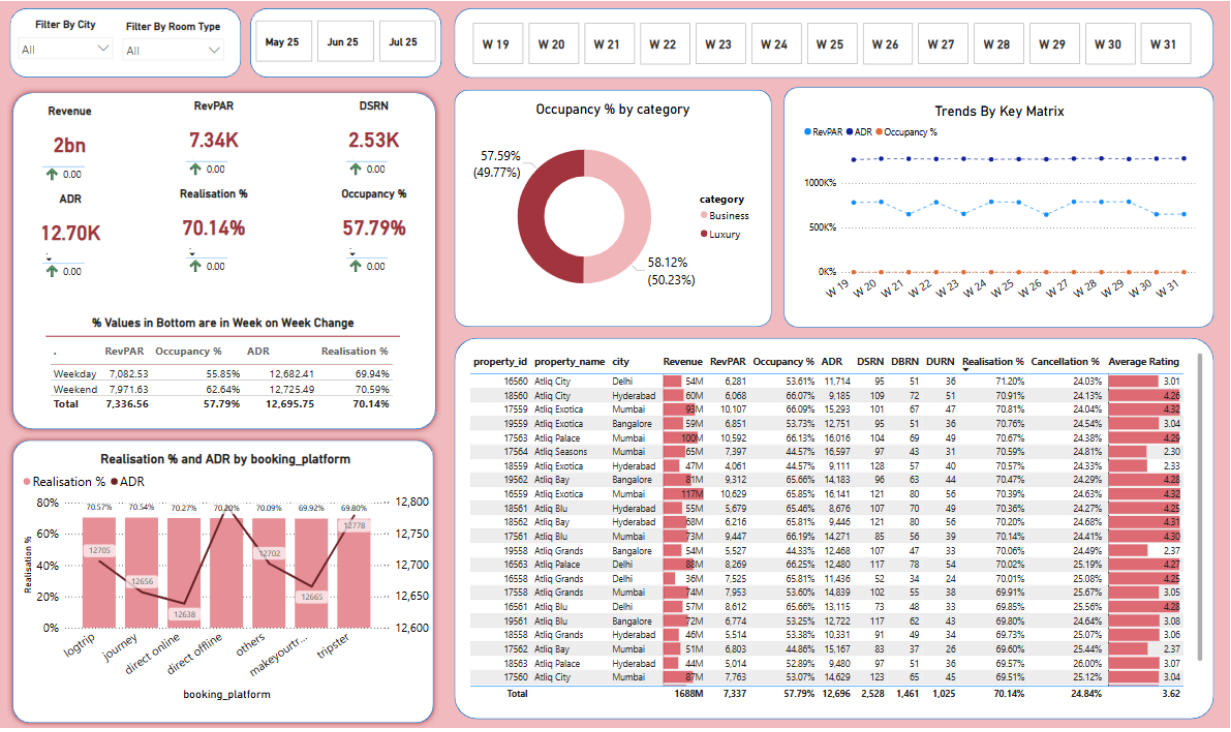
```
Revenue = SUM(fact_booking[revenue_realized])
Total Successful Bookings = SUM(fact_aggregated_bookings[successful_bookings])
```

Measures are crucial since they allow one to derive meaningful insights by applying dynamic aggregations, calculations, and business logic across datasets. Unlike calculated columns, which generate static values at the row level, measures usually provide flexibility by computing results based on the user interactions, such as filtering.

Here are some of the columns in the **key_measures_table**:


key_measures_table
ADR
ADR WoW change %
Average Rating
Booking % by platform
Booking % by Room class
Cancellation %
DBRN
DSRN
DSRN WoW change %
DURN
No of Days
No Show rate %
Occupancy %
Occupancy WoW change %
Realisation %
Realisation WoW change %
Revenue
Revenue WoW change %
RevPAR
Revpar WoW change %
Total Bookings
Total Cancelled Bookings

Step 5: Creating a Visualization Dashboard



.	RevPAR	Occupancy %	ADR	Realisation %
Weekday	7,082.53	55.85%	12,682.41	69.94%
Weekend	7,971.63	62.64%	12,725.49	70.59%
Total	7,336.56	57.79%	12,695.75	70.14%

property_id	property_name	city	Revenue	RevPAR	Occupancy %	ADR	DSRN	DBRN	DURN	Realisation %	Cancellation %	Average Rating
16560	Atliq City	Delhi	54M	6,281	53.61%	11,714	95	51	36	71.20%	24.03%	3.01
18560	Atliq City	Hyderabad	60M	6,068	66.07%	9,185	109	72	51	70.91%	24.13%	4.26
17559	Atliq Exotica	Mumbai	98M	10,107	66.09%	15,293	101	67	47	70.81%	24.04%	4.32
19559	Atliq Exotica	Bangalore	59M	6,851	53.73%	12,751	95	51	36	70.76%	24.54%	3.04
17563	Atliq Palace	Mumbai	100M	10,592	66.13%	16,016	104	69	49	70.67%	24.38%	4.29
17564	Atliq Seasons	Mumbai	65M	7,397	44.57%	16,597	97	43	31	70.59%	24.81%	2.30
18559	Atliq Exotica	Hyderabad	47M	4,061	44.57%	9,111	128	57	40	70.57%	24.33%	2.33
19562	Atliq Bay	Bangalore	81M	9,312	65.66%	14,183	96	63	44	70.47%	24.29%	4.28
16559	Atliq Exotica	Mumbai	117M	10,629	65.85%	16,141	121	80	56	70.39%	24.63%	4.32
18561	Atliq Blu	Hyderabad	55M	5,679	65.46%	8,676	107	70	49	70.36%	24.27%	4.25
18562	Atliq Bay	Hyderabad	68M	6,216	65.81%	9,446	121	80	56	70.20%	24.68%	4.31
17561	Atliq Blu	Mumbai	73M	9,447	66.19%	14,271	85	56	39	70.14%	24.41%	4.30
19558	Atliq Grands	Bangalore	54M	5,527	44.33%	12,468	107	47	33	70.06%	24.49%	2.37
16563	Atliq Palace	Delhi	88M	8,269	66.25%	12,480	117	78	54	70.02%	25.19%	4.27
16558	Atliq Grands	Delhi	36M	7,525	65.81%	11,436	52	34	24	70.01%	25.08%	4.25
17558	Atliq Grands	Mumbai	74M	7,953	53.60%	14,839	102	55	38	69.91%	25.67%	3.05
16561	Atliq Blu	Delhi	57M	8,612	65.66%	13,115	73	48	33	69.85%	25.56%	4.28
19561	Atliq Blu	Bangalore	72M	6,774	53.25%	12,722	117	62	43	69.80%	24.64%	3.08
18558	Atliq Grands	Hyderabad	46M	5,514	53.38%	10,331	91	49	34	69.73%	25.07%	3.06
17562	Atliq Bay	Mumbai	51M	6,803	44.86%	15,167	83	37	26	69.60%	25.44%	2.37
18563	Atliq Palace	Hyderabad	44M	5,014	52.89%	9,480	97	51	36	69.57%	26.00%	3.07
17560	Atliq City	Mumbai	87M	7,763	53.07%	14,629	123	65	45	69.51%	25.12%	3.04
Total			1688M	7,337	57.79%	12,696	2,528	1,461	1,025	70.14%	24.84%	3.62

This is the final dashboard I managed to come up with.

- It has filters sections.
- Filter by city where one can filter the data based on the city.
- Filter by Room Type where one can filter based on the room type like maybe “presidential”.
- One can filter also based on the “months” of May, June, and July.
- Alternatively, also filter based on the week number.
- It has KPIs like “Revenue”, “RevPAR”, “DSRN”, “ADR”, “Realisation %” and “Occupancy %” which change dynamically when one applies the filters.

Link to Dashboard.

I didn’t manage to share it through the Power BI page.

I created a folder in my one drive account and uploaded the pdf version of the dashboard, the screenshot and the final Power BI file.

I then made then link accessible by everyone and shared it here:

Link to Power BI files:

Dashboard Files.

Conclusion

Through this hands-on experience with Power BI, I have gained a deeper understanding of how data analysis can drive informed decision-making in real world applications like in hotel management. From transforming datasets to building measures and crafting insightful visualizations, each step has reinforced the importance of data-driven strategies in business intelligence. The final dashboard not only showcases key performance indicators but also serves as a powerful tool for optimizing operations. This assignment has strengthened my analytical skills and demonstrated the real-world impact of business intelligence, which I also believe that it has prepared me for more advanced applications in the future.